Section 1: Implementation Logic

System Prompt Strategy:

Paste your System Prompt. Highlight the Few-Shot Example you added. Explain why this example helps the model.

Full prompt:

```
SYSTEM_PROMPT = '''
You are a helpful assistant who can do reasoning and acting to give the
user a reasonable answer.
You are allowed to use the tools to help you get the sufficient
information to answer the query.
You must give your thought and action following the format below and only
one thought and action per every step, the user will help you count the
step:
user: [step n]
Thought: <Breaking the query and thinking of what to do and how to do.>
Action: [{"name": "tool_name", "args": {"arg1":"arg1", "arg2": "arg2",
...}},...] (The tool you want to use, json format. Can use multiple tools
if you need.)>

Then you will get the informations:
Observation: <the information> (Don't generate observations yourself!)
If you get enough informations, then generate the answer to the query:
Thought: I've gotten enough information to answer.
Action: [{"name": "answer", "args": {"answer":"final answer"}}]

For example:
user query: "How to make cake?"
user: [step 1]
Thought: I need to search "How to make cake?" on the web to answer the
query.
Action: [{"name": "search_web", "args": {"query": "How to make cake?"}}]
Observation:
search_web "How to make cake?"
How to make cake? First, You need to prepare...

user: [step 2]
Thought: I've gotten enough information to answer.
Action: [{"name": "answer", "args": {"answer":"To make cake, you need to
prepare..."}}]
```

```
_____________________________
The tools you can use, the tool name and the arg name must be same as the
description:
1. answer: stop ReAct and answer the query
    args: {"answer": "the final answer"}

2. search_web: Gives the query that you want to search on the web. Return
json list of the search results.
    args: {"query": "The query that you want to search on the web"}

3. calculate: Do math calculation
    args: {"expression": "math expression, e.g.: (3 + 5) * 2"}
_____________________________
Don't respond without enough information.
Don't use the tool that isn't on the tool list.
'''
```

1. Avoid acting twice in one step and give format.

```
You must give your thought and action following the format below and only
one thought and action per every step, the user will help you count the
step:
user: [step n]
Thought: <Breaking the query and thinking of what to do and how to do.>
Action: [{"name": "tool_name", "args": {"arg1":"arg1", "arg2": "arg2",
...}},...] (The tool you want to use, json format. Can use multiple tools
if you need.)>
```

The code below is about appending the counting step message per every step.

```
self.messages.append({"role": "user", "content": f"[step {iteration+1}]"})
```

2. Give an example of multi-step.

```
For example:
user query: "How to make cake?"
user: [step 1]
Thought: I need to search "How to make cake?" on the web to answer the
query.
Action: [{"name": "search_web", "args": {"query": "How to make cake?"}}]
Observation:
search_web "How to make cake?"
How to make cake? First, You need to prepare...

user: [step 2]
Thought: I've gotten enough information to answer.
Action: [{"name": "answer", "args": {"answer":"To make cake, you need to
prepare..."}}]
```

3. Give tools and the description.

```
_____________________________
The tools you can use, the tool name and the arg name must be same as the
description:
1. answer: stop ReAct and answer the query
    args: {"answer": "the final answer"}

2. search_web: Gives the query that you want to search on the web. Return
json list of the search results.
    args: {"query": "The query that you want to search on the web"}

3. calculate: Do math calculation
    args: {"expression": "math expression, e.g.: (3 + 5) * 2"}
_____________________________
```

The Loop Mechanism:

Briefly explain how your Python script feeds the "Observation" back into the LLM's context window

1. The model generates thought and action to try to solve the query.
2. Using parser to parse the thought and action, the action follows the format below:

```
"actions": [
    {
        "name": "tool_1",
        "args": {
            "arg1": "arg1",
            "arg2": "arg2",
            ...
        }
    },
    ...
]
```

3. Map the tools and arguments, get the result of function calling, add in observation text, and the observation text will append in messages.(feed back into LLM's context)

```
# tools.py
TOOLS: dict[str, callable] = {
    'search_web':    lambda inp: search_web(inp['query']),
    'calculate':     lambda inp: calculate(inp['expression']),
}
```

agent.py

```
# Call Tool ----------------
observation_parts = ["Observation:"]
for action in parsed['actions']:
    tool_name = action['name']
    args = action["args"]
    if tool_name not in TOOLS:
        error_msg = f"Tool '{tool_name}' not found."
        self._log_error(error_msg)
        break
    tool_result = TOOLS[tool_name](args)
    if 'error' in tool_result:
        error_msg = f"Fail to call tool: {action}\n{tool_result['error']}"
        self._log_error(error_msg)
        break
    try:
        observation_parts.append(f"{tool_name}({','.join([str(v) for v in args.values()])})")
        observation_parts.append(str(tool_result))
    except Exception as e:
        error_msg = f"Fail to call tool: {action}\n{e}"
        self._log_error(error_msg)
        break
observation_text = '\n'.join(observation_parts)
```

Section 2: Benchmark Traces (The Evidence)

For each task, paste the Console Trace and analyze the specific behavior.

Task 1: Planning & Quantitative Reasoning

Question: "What fraction of Japan's population is Taiwan's population as of 2025?"

1. Step1: LLM knows to get the population and do calculations. Call search_web to get Japan's population.

```
========== Step 1 ==========
Thought: I need to get the current populations of Japan and Taiwan in 2025 first, then calculate the fraction.
Action: [{'name': 'search_web', 'args': {'query': 'Japan population 2025'}}]

Please wait for the search result of Japan's population 2025.
--------------------
Observation:
search_web(Japan population 2025)
title: Japan Population, 1899 - 2025 | CEIC Data
url: https://www.ceicdata.com/en/indicator/japan/population
content: Japan Population reached 123.2 million people in Dec 2025, compared with the previously reported figure of 123.7 million people in Dec 202
4.

title: Population of Japan 2025 - PopulationPyramid.net
url: https://www.populationpyramid.net/japan/2025/
content: Population pyramid of Japan in 2025. Total population: 123103479. Interactive age and sex distribution visualization from 1950 to 2100.

title: Japan Population (1950-2025) - Macrotrends
url: https://www.macrotrends.net/global-metrics/countries/jpn/japan/population
content: ## Japan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Crime Immigration Other. Po
pulation Growth Rate Density Urban Rural Life Expectancy Birth Rate Death Rate Infant Mortality Rate Fertility Rate. ##### Total current population
 for Japan in 2025 was **121,960,408**, a **0.55% decline** from 2024. * Total population for Japan in 2024 was **122,631,432**, a **1.51% decline*
* from 2023. * Total population for Japan in 2023 was **124,516,650**, a **0.49% decline** from 2022. * Total population for Japan in 2022 was **12
5,124,989**, a **0.44% decline** from 2021. Total population is based on the de facto definition of population, which counts all residents regardle
ss of legal status or citizenship. The values shown are midyear estimates. The full historical dataset is available for download here: Population o
f Japan | Historical Data | 1950 - 2025.
--------------------
```

2. Step2: Call search_web to get Taiwan population.

```
========== Step 2 ==========
Thought: I've obtained the estimated population of Japan in 2025, but I still need to get the actual population of Taiwan in 2025.
Action: [{'name': 'search_web', 'args': {'query': 'Taiwan population 2025'}}]
--------------------
Observation:
search_web(Taiwan population 2025)
title: Taiwan demographics 2026 - StatisticsTimes.com
url: https://statisticstimes.com/demographics/country/taiwan-demographics.php
content: As of 2026, Taiwan's population is projected at 23.01 million, compared to 23.11 million in 2025. It ranks 61st in the world by population

title: Taiwan Population (1950-2025) - Macrotrends
url: https://www.macrotrends.net/global-metrics/countries/twn/taiwan/population
content: ## Taiwan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Crime Immigration Other. P
opulation Growth Rate Density Urban Rural Life Expectancy Birth Rate Death Rate Infant Mortality Rate Fertility Rate. ##### Total current populatio
n for Taiwan in 2025 was **23,974,031**, a **0.1% increase** from 2024. * Total population for Taiwan in 2024 was **23,950,214**, a **0.11% increas
e** from 2023. * Total population for Taiwan in 2023 was **23,923,276**, a **0.13% increase** from 2022. * Total population for Taiwan in 2022 was
**23,893,394**, a **0.14% increase** from 2021. Total population is based on the de facto definition of population, which counts all residents rega
rdless of legal status or citizenship. The values shown are midyear estimates. The full historical dataset is available for download here: Populati
on of Taiwan | Historical Data | 1950 - 2025.

title: Taiwan population 2025: 23.1M 2096: 10.5M (projected) (World ...
url: https://x.com/Thefactualdata/status/1954365570595377178
content: The Factual Data. Thefactualdata. Aug 9. Taiwan population 2025: 23.1M 2096: 10.5M (projected) (World Population Review).
--------------------
```

3. Step3: Calculate the fraction.

```
========== Step 3 ==========
Thought: I've obtained the population of Japan and Taiwan in 2025, now I can calculate the fraction of Japan's population that is Taiwan's populati
on.
Action: [{'name': 'calculate', 'args': {'expression': '23470000 / 123103479'}}]
--------------------
Observation:
calculate(23470000 / 123103479)
23470000 / 123103479 = 0.19065261348137855
--------------------
```

4. Step4: Give the answer “19%”

```
========== Step 4 ==========
Thought: I've gotten the fraction of Japan's population that is Taiwan's population, now I can convert it to a fraction.
Action: [{'name': 'answer', 'args': {'answer': '190/1000 or 19%'}}]
Answer:
190/1000 or 19%
```

Task 2: Technical Specificity

Question: "Compare the main display specs of iPhone 15 and Samsung S24."

1. Step1: Search for general display specs of iPhone 15 and Samsung Galaxy S24.

```
========== Step 1 ==========
Thought: To compare the main display specs of iPhone 15 and Samsung S24, I need to gather information about their display specifications from relia
ble sources.
Action: [{"name": "search_web", "args": {"query": "iPhone 15 display specs vs Samsung S24 display specs"}}]
--------------------
Observation:
search_web(iPhone 15 display specs vs Samsung S24 display specs)
title: Samsung Galaxy S24 vs. Apple iPhone 15: Which Flagship ...
url: https://me.pcmag.com/en/mobile-phones/21583/samsung-galaxy-s24-vs-apple-iphone-15-which-flagship-phone-is-best
content: Samsung sells three Galaxy S24 devices, while Apple offers four iPhone 15 models. Similarly, Apple's top-of-the-line iPhone 15 Pro Max cos
ts $100 less than the Galaxy S24 Ultra. Samsung's Galaxy S24+ measures 6.2 by 3.0 by 0.30 inches and weighs 6.95 ounces, making it smaller and ligh
ter than the iPhone 15 Plus (6.33 by 3.06 by 0.31 inches, 7.09 ounces) and the iPhone 15 Pro Max (6.29 by 3.02 by 0.32 inches, 7.81 ounces). Left t
o right: Galaxy S24 Ultra, iPhone 15 Pro Max. The S24 has a 6.2-inch display with FHD+ resolution (2,340 by 1,080 pixels), while the 6.7-inch S24+
and 6.8-inch S24 Ultra phones both have QHD+ (3,120 by 1,440 pixels) displays. Left to right: Galaxy S24 Ultra, iPhone 15 Pro Max. The Pro models g
et three rear cameras: a 48MP f/1.78 main lens with OIS, a 12MP f/2.2 ultra-wide camera with a 120-degree field of view, and a telephoto option. Mo
reover, the Apple Watch and AirTag work only with iPhones, while the Galaxy Watch and SmartTag 2 perform best with Samsung phones.

title: Samsung Galaxy S24 vs iPhone 15
url: https://www.pocket-lint.com/samsung-galaxy-s24-vs-iphone-15/
content: # Samsung Galaxy S24 vs iPhone 15. * Galaxy S24 and iPhone 15 are similarly priced at $799, but Samsung offers deals, giving more value. *
 Samsung Galaxy S24 has an edge over iPhone 15 with its 120Hz display, adaptive refresh rate, and better camera options. * | Samsung Galaxy S24 | A
pple iPhone 15 |. That’s why both Samsung and Apple take this field extremely seriously, and both the Galaxy S24 and the iPhone 15 are a great choi
ce if you rely on your phone for taking photos. These phones come with great battery lives -- the iPhone 15 does better here, offering incredible p
erformance out of its relatively small 3349mAh cell, but Samsung is also doing great, offering whole-day use (especially with a more efficient SoC
and screen) out of the 4000mAh battery. Apple and Samsung's newest handsets use a USB-C charging port, but both max out at 25W wired charging, with
 15W wireless charging, which is not that fast -- even compared to Samsung's Galaxy S24 Ultra which features 65W wired charging.

title: Apple iPhone 15 vs Samsung Galaxy S24 - Comparison
url: https://www.phonearena.com/phones/compare/Apple-iPhone-15,Samsung-Galaxy-S24/phones/11927,12113
content: # Apple iPhone 15 vs Samsung Galaxy S24 Specs Comparison. | 48 MP (Sensor-shift OIS, PDAF) Sensor name: Sony IMX904 Aperture size: F1.6 Fo
cal length: 26 mm Pixel size: 2.0 µm | 50 MP (OIS, PDAF) Sensor name: Samsung GN3 Aperture size: F1.8 Focal length: 24 mm Sensor size: 1/1.56" Pixe
l size: 1.0 µm |. | 12 MP (Ultra-wide) Sensor name: Sony IMX772 Aperture size: F2.4 Focal Length: 13 mm 12 MP (Ultra-wide) Sensor name: Sony IMX564
 Aperture size: F2.0 Focal Length: 13 mm Sensor size: 1/2.55" Pixel size: 1.4 µm | |. |  | 10 MP (Telephoto) Sensor name: Samsung S5K3K1 Aperture s
ize: F2.4 Focal Length: 67 mm Sensor size: 1/3.94" Pixel size: 1.0 µm |. | | Full Specs | | --- | | Apple iPhone 15 Specs | | PA Review | | Apple i
Phone 15 Review | | | Full Specs | | --- | | Samsung Galaxy S24 Specs | | PA Review | | Samsung Galaxy S24 Review | |.
--------------------
```

2. Step2: Look specifically for display resolution details to refine the comparison.

```
========== Step 2 ==========
Thought: I've gathered some information about the display specifications of iPhone 15 and Samsung S24, but I still need more specific data to accur
ately compare them.
Action: [{"name": "search_web", "args": {"query": "iPhone 15 display resolution vs Samsung S24 display resolution"}}]
--------------------
Observation:
search_web(iPhone 15 display resolution vs Samsung S24 display resolution)
title: Resolution - Apple iPhone 15 vs Samsung Galaxy S24 - Versus
url: https://versus.com/en/apple-iphone-15-vs-samsung-galaxy-s24/resolution
content: # Resolution. Resolution is an essential indicator of a screen's image quality, representing the maximum amount of pixels that can be show
n on the screen. The resolution is given as a compound value, comprised of horizontal and vertical pixels. Resolution: is this important to you? Th
is page is currently only available in English.

title: Samsung Galaxy S24 vs. Apple iPhone 15: Which Flagship Phone Is ...
url: https://me.pcmag.com/en/mobile-phones/21583/samsung-galaxy-s24-vs-apple-iphone-15-which-flagship-phone-is-best
content: Samsung sells three Galaxy S24 devices, while Apple offers four iPhone 15 models. Similarly, Apple's top-of-the-line iPhone 15 Pro Max cos
ts $100 less than the Galaxy S24 Ultra. Samsung's Galaxy S24+ measures 6.2 by 3.0 by 0.30 inches and weighs 6.95 ounces, making it smaller and ligh
ter than the iPhone 15 Plus (6.33 by 3.06 by 0.31 inches, 7.09 ounces) and the iPhone 15 Pro Max (6.29 by 3.02 by 0.32 inches, 7.81 ounces). Left t
o right: Galaxy S24 Ultra, iPhone 15 Pro Max. The S24 has a 6.2-inch display with FHD+ resolution (2,340 by 1,080 pixels), while the 6.7-inch S24+
and 6.8-inch S24 Ultra phones both have QHD+ (3,120 by 1,440 pixels) displays. Left to right: Galaxy S24 Ultra, iPhone 15 Pro Max. The Pro models g
et three rear cameras: a 48MP f/1.78 main lens with OIS, a 12MP f/2.2 ultra-wide camera with a 120-degree field of view, and a telephoto option. Mo
reover, the Apple Watch and AirTag work only with iPhones, while the Galaxy Watch and SmartTag 2 perform best with Samsung phones.

title: Samsung Galaxy S24 vs. Apple iPhone 15 - Android Central
url: https://www.androidcentral.com/phones/samsung-galaxy-s24-vs-apple-iphone-15
content: Samsung is marketing the Galaxy S24 as an "AI phone," which distracts from the fact that it's *very* similar to the S23 aside from some di
splay upgrades and the promise of longer software support. The iPhone 15 remains stuck at 60Hz, and iMore notes that the stuttering difference is e
specially noticeable when "using menus and unlocking the phone." Meanwhile, the Galaxy S24 runs silky smooth at 120Hz. Hands-on with the smaller Sa
msung Galaxy S24. | Category | Samsung Galaxy S24 | Apple iPhone 15 |. Because of that, the benchmarks for the iPhone 15 and Galaxy S24 should be f
airly comparable, except that Apple's phone should perform faster for straightforward single-core tasks, and Samsung's phone will *probably* perfor
m better for gaming or creativity apps. The Galaxy S24 is being marketed as an "AI phone" with a revamped NPU to handle things like live translatio
n; Apple, meanwhile, gave the iPhone 15 a new "16-core Neural Engine" that's apparently "capable of nearly 17 trillion operations per second," with
 functions like live transcriptions of voicemail.
--------------------
```

3. Step3: Gather information about display size differences between the two phones.

```
========== Step 3 ==========
Thought: I need to gather more specific information about the display specs of Samsung S24.
Action: [{"name": "search_web", "args": {"query": "Samsung S24 display specs"}}]
--------------------
Observation:
search_web(Samsung S24 display specs)
title: Samsung S24 series tech specs (S24, S24+, S24 FE, S24 Ultra)
url: https://www.backmarket.com/en-us/c/samsung/samsung-s24-tech-specs?srsltid=AfmBOorfPfy2VwfJwmtIw_7GPNzHQRHQPSMe0LOJIgmYXowRE6bj8JWu
content: # Samsung S24 series tech specs (S24, S24+, S24 FE, S24 Ultra). We're here to give you more information on the Samsung S24 series release
date, camera specs, and overall features. Under the hood, the Galaxy S24+ is powered by a Qualcomm Snapdragon 8 Gen 3 processor, and other specs in
clude a triple camera setup and a 4,900mAh battery. ## Samsung Galaxy S24 FE specs. The Samsung Galaxy S24 FE introduces a 6.7-inch Dynamic AMOLED
display with a smooth 120Hz refresh rate, with the Exynos 2400e processor under the hood. ## Samsung Galaxy S24 Ultra specs. The Samsung Galaxy S24
 Ultra sits at the top of the pyramid with a 6.8-inch Dynamic AMOLED display with a 120Hz refresh rate. The Samsung S24 Ultra features a 5,000mAh b
attery, and its camera specs include a quad camera setup on the back. + **Video capture:** 3840x2160 (4K UHD) (60 fps). You can purchase a Samsung
S24, S24+, S24 FE, or S24 Ultra new or refurbished.

title: Samsung Galaxy S24 - Specifications
url: https://www.devicespecifications.com/en/model/20d85e13
content: Storage: 128 GB, 256 GB, 512 GB ; Display: 6.2 in, Dynamic AMOLED 2X, 1080 x 2340 pixels, 24 bit ; Battery: 4000 mAh, Li-Polymer ; OS: And
roid 14 ; Camera: 50 MP,

title: Samsung Galaxy S24 Full Specifications
url: https://www.phonearena.com/phones/Samsung-Galaxy-S24_id12113
content: Size, 6.2-inch, 90.87% screen-to-body ; Type, Dynamic AMOLED, 120Hz, HDR, 2600 nits ; Resolution, 2340x1080px, 19.5:9 ratio, 416 PPI ; Pro
tection
--------------------
```

4. Step4: Find side-by-side comparisons or reviews for more complete and reliable data.

```
========== Step 4 ==========
Thought: I still need more information to accurately compare the display specifications of iPhone 15 and Samsung S24. I should find a review that s
pecifically compares these two phones.
Action: [{"name": "search_web", "args": {"query": "side-by-side comparison of iPhone 15 and Samsung S24 displays"}}]
--------------------
Observation:
search_web(side-by-side comparison of iPhone 15 and Samsung S24 displays)
title: iPhone 15 series vs. Samsung S24 series comparison - Back Market
url: https://www.backmarket.com/en-us/c/samsung/samsung-s24-series-vs-iphone-15-series-comparison?srsltid=AfmBOooUAVSEdxRLIt8aPZBYCHp1R2LpHqprwColY
pBZ4LeUOYnGLWVV
content: **The Galaxy S24 stands out as a more innovative smartphone thanks to its Galaxy AI features**, and it also offers several advantages over
 the iPhone base models, like faster charging, a higher refresh rate, and its triple-camera system. The Samsung Galaxy S24+ and the iPhone 15 Pro b
oth offer a step up from their respective base models in processing power and RAM, making their performance even better. **The iPhone 15 Pro rivals
 the Galaxy S24+ in ways that its base model siblings don't: it gets the same high screen refresh rate of 120Hz for smooth scrolling and also joins
 the S24+ with a triple-camera setup.** Finally, the 15 Pro incorporates **Apple Intelligence**, Apple's answer to Galaxy AI. The Samsung Galaxy S2
4 Ultra slightly wins out over the iPhone 15 Pro Max when it comes to the top-tier models thanks to its expanded storage capacity, its quad-lens ca
mera upgrade, and its support of the S Pen removable stylus.

title: Samsung Galaxy S24 vs iPhone 15: the key differences - TechRadar
url: https://www.techradar.com/phones/samsung-galaxy-s24-vs-iphone-15
content: iPhone 3GS vs iPhone 17 Pro Max. Samsung Galaxy S26 Ultra REVIEW. We pit the Samsung Galaxy S24 against the iPhone 15. ## Samsung Galaxy S
24 vs iPhone 15: Specs comparison. Samsung Galaxy S24 vs iPhone 15: Specs. |  | iPhone 15 | Samsung Galaxy S24 |. ## Samsung Galaxy S24 vs iPhone 1
5: Design and display. ## Samsung Galaxy S24 vs iPhone 15: Cameras. So the Samsung Galaxy S24 benefits from an extra camera, albeit not that much m
ore versatility once you factor in the iPhone 15's 2x optical quality crop. ## Samsung Galaxy S24 vs iPhone 15: Performance. ## Samsung Galaxy S24
vs iPhone 15: Verdict. A Samsung Galaxy S24 and an iPhone 15. A Samsung Galaxy S24 and an iPhone 15. The Galaxy S24, however, benefits from a varia
ble refresh rate (which the iPhone 15 doesn't have), as well as slightly more versatile cameras (you get a whole telephoto lens with the Samsung ph
one).

title: Hands-on: How does the Galaxy S24 compare to the iPhone 15?
url: https://9to5mac.com/2024/01/23/hands-on-galaxy-s24-compare-iphone-15/
content: Currently, my main smartphone is the iPhone 15 Pro Max. Since last week, I've been using the Galaxy S24 to try out the new features (which
 are mostly AI-related) and challenge myself to use a phone that isn't huge. In recent years, Samsung has come up with some pretty original designs
, but I have to say that the Galaxy S24 looks a lot like the latest iPhone models. The new phones have a faster SoC and slightly better displays, b
ut what sets them apart from the Galaxy S23 models are the AI-based features that come built-in. And again, Samsung isn't introducing anything new
– but having these features built into the phone will make a difference to a lot of users who would never download third-party AI tools. I will con
tinue to use the iPhone as my main phone for many reasons (especially because of the Apple ecosystem), but I have to admit that using the Galaxy S2
4 doesn't make me miss many things about my iPhone.
--------------------
```

5. Step 5: Summarize the collected information into a concise comparison of display specs and give an answer.

```
========== Step 5 ==========
Thought: Now that I have a better understanding of the display specifications of iPhone 15 and Samsung S24, I need to summarize the comparison in a
 clear and concise manner.
Action: [{"name": "answer", "args": {"answer": "The display specifications of iPhone 15 and Samsung S24 are similar, with both phones offering high
-resolution displays and fast refresh rates. However, the Samsung S24 has a slight edge in terms of display size and resolution, with the S24+ mode
l offering a 6.7-inch display with QHD+ resolution (3,120 by 1,440 pixels), compared to the iPhone 15's 6.1-inch display with FHD+ resolution (2,34
0 by 1,080 pixels). Additionally, the Samsung S24 supports a variable refresh rate, which can help to improve battery life and reduce eye strain."}
}]
Answer:
The display specifications of iPhone 15 and Samsung S24 are similar, with both phones offering high-resolution displays and fast refresh rates. How
ever, the Samsung S24 has a slight edge in terms of display size and resolution, with the S24+ model offering a 6.7-inch display with QHD+ resoluti
on (3,120 by 1,440 pixels), compared to the iPhone 15's 6.1-inch display with FHD+ resolution (2,340 by 1,080 pixels). Additionally, the Samsung S2
4 supports a variable refresh rate, which can help to improve battery life and reduce eye strain.
```

Task 3: Resilience & Reflection Test

Question: "Who is the CEO of the startup 'Morphic' AI search?"

1. Step1: Use “search_web” to search “ Morphic' AI search CEO"

```
========== Step 1 ==========
Thought: I need to search "Morphic AI search CEO" on the web to find who is the CEO of Morphic AI search.
Action: [{"name": "search_web", "args": {"query": "Morphic AI search CEO"}}]
--------------------
Observation:
search_web(Morphic AI search CEO)
title: Morphic holding CEO sells shares worth over $1.2 million
url: https://ng.investing.com/news/company-news/morphic-holding-ceo-sells-shares-worth-over-12-million-93CH-1434356
content: (NASDAQ:MORF) President and CEO, Praveen P. Tipirneni, has made significant transactions in the company's stock, according to a recent SEC

title: "Harness those cross-functional skills" with Weston Baker of Morphic
url: https://www.youtube.com/watch?v=daorSuHJNpg
content: Weston Baker, Founder & CEO of Morphic and an overall force in the ... ai/blog/what-the-tech-episode-21-harness-cross-functional-skills

title: Morphic - 2026 Company Profile & Team - Tracxn
url: https://tracxn.com/d/companies/morphic/__QREWaDzuHM-XW-PFwSgyqKupVzTD3aOF5VD9L0I9mSw
content: Your browser was unable to load all of Tracxn resources. # Morphic - Company Profile. ## Morphic - About the company. Morphic is an unfund
ed company based in San Francisco (United States), foundedbyJaynti Kanani. Morphic has not raised any funding yet. View Morphic's full profile. Sig
n up to download Morphic's company profile. ## Morphic's funding and investors. Morphic has not raised any funding rounds yet. ## Morphic's founder
s and board of directors. * Jaynti Kanani: Founder & CEO of Morphic and founder of 1 other company. ## Morphic's Investments and acquisitions. ## R
eports related to Morphic. Viewall reports related to Morphic. View complete company profile of Morphic. ## FAQs about Morphic. * GAYA BAZAAR - Gay
a based, 2023 founded, Unfunded company. * Nexogenius Technologies - Rajkot based, 2023 founded, Unfunded company. * Xcelsior Data Services - Puduc
herry based, 2022 founded, Unfunded company. * Ptc Shop - Australia based, 2009 founded, Unfunded company. * TCI Chemicals - Tokyo based, 1946 foun
ded, Unfunded company.
--------------------
```

2. Step2: Use “search_web” to verify the CEO.

```
========== Step 2 ==========
Thought: I need to verify if Weston Baker or Praveen P. Tipirneni is actually the CEO of Morphic AI search.
Action: [{"name": "search_web", "args": {"query": "Morphic AI search CEO Weston Baker Praveen P. Tipirneni"}}]
--------------------
Observation:
search_web(Morphic AI search CEO Weston Baker Praveen P. Tipirneni)
title: Founder Stories: Praveen Tipirneni, CEO of Morphic Therapeutic
url: https://www.youtube.com/watch?v=d3vYYfhDsqU
content: Founder Stories: Praveen Tipirneni, CEO of Morphic Therapeutic
Axial
4810 subscribers
3 likes
475 views
25 May 2023
Praveen Tipirneni, MD, MBA is Chief Executive Officer of Morphic Therapeutic Inc. Previously, he was Senior Vice President of Corporate Development
 and Global Strategy at Cubist Pharmaceuticals, a position in which he served from 2002 until the company’s acquisition by Merck in 2015. In his ti
me at Cubist, he was a member of the clinical group working on the Cubicin NDA (skin and skin structure infections) and sNDA (Staph. Bacteremia and
 Endocarditis) teams. He was head of business development since January 2006. Prior to joining Cubist, Dr. Tipirneni worked at Sun Microsystems in
corporate strategy, Covad Communications in Corporate Strategy, and Deltagen in business development. He also served time as a 1st Lieutenant in th
e U.S. Army. Dr. Tipirneni received a bachelor’s degree from MIT in mechanical engineering and an M. D. from McGill University. After completing hi
s post-graduate residency in Internal Medicine at University of Illinois, Chicago, he received his MBA from the University of Pennsylvania’s Wharto
n School of Business in healthcare finance.

https://www.linkedin.com/in/praveen-tipirneni-42a1a5

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title: Biotech 2050 Podcast | Podcast on Podbay
url: https://podbay.fm/p/biotech-2050-podcast
content: Biography: Praveen Tipirneni, MD, MBA is Chief Executive Officer of Morphic Therapeutic Inc. Previously, he was Senior Vice President of C
orporate Development and Global Strategy at Cubist Pharmaceuticals, a position in which he served from 2002 until the company’s acquisition by Merc
k in 2015. Synopsis: In this episode of Biotech2050, Dr. Ted Love, former CEO of Global Blood Therapeutics and Current Chairman of BIO, shares insi
ghts on sickle cell breakthroughs, transitioning from academia to research, navigating a tight funding environment, and protecting an innovative ec
osystem to benefit patients. Biography: Scott joined Adiso Therapeutics as Chief Executive Officer in January 2022, bringing more than 35 years of
successful industry leadership, having held multiple senior executive, CEO and board director roles during his career at both Fortune 500 pharmaceu
tical and emerging biotechnology companies. Synopsis: In this episode of the Biotech2050 podcast, Dr. Mathai Mammen, CEO of FogPharma, shares valua
ble insights into the future trends of the biotech industry and the development of peptide-based therapeutics.
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3. Step3: Answer: Praveen Tipirneni is the CEO of Morphic Therapeutic.

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========== Step 3 ==========
Thought: I need to determine which of Weston Baker or Praveen P. Tipirneni is the current CEO of Morphic AI search.
Action: [{"name": "answer", "args": {"answer": "Praveen Tipirneni is the CEO of Morphic Therapeutic Inc, but I am unable to verify that Morphic The
rapeutic Inc is Morphic AI search.  A search was made and did not find Morphic AI search, Morphic AI (AI is a field and not a company, and Morphic
search AI  search may be a startup but this is not what came up as Morphic search AI  so I am uncertain which is current company"}}]
Answer:
Praveen Tipirneni is the CEO of Morphic Therapeutic Inc, but I am unable to verify that Morphic Therapeutic Inc is Morphic AI search.  A search was
 made and did not find Morphic AI search, Morphic AI (AI is a field and not a company, and Morphic search AI  search may be a startup but this is n
ot what came up as Morphic search AI  so I am uncertain which is current company
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